

Should You Use Power Boosted or Manual Brakes on Your Next Project?

Many popular hobby cars come from the era when power brakes were an extra-cost option, and drum brakes were the norm. While Wilwood prefers manual brake systems, a holdover from our racing roots, our calipers and master cylinders work equally well for power or manual brakes. If you are just planning your build, now is the time to decide if you are going to go with boosted brakes or not.

When planning a performance manual brake system, such as for a race car, muscle car or street rod, getting the pedal and hydraulic ratio correct is even more critical. The hydraulic ratio relates the master cylinder piston size to the piston area in the calipers. The pedal ratio is the relation of the pedal pad, master cylinder pushrod, and brake pedal pivot. Both amplify your muscle effort to help the car stop, with or without a power booster. Get them both right, and even with manual brakes, you'll be able to stop your car or truck on the proverbial dime.

Power Assist vs. Manual Brakes

For some projects, the question of power assist or manual brakes is merely a personal preference, but the choice is forced on you for others. For example, often shoehorning a big motor into a small engine compartment leaves no room for a vacuum booster on the firewall, even with an angled (Ford Bronco) or offset (many vintage vehicles) mounting bracket. Other times, the radical cam timing of a high output motor doesn't create enough vacuum to power a traditional vacuum booster.

Manual Brakes



Because you need to be more precise when racing, and every datum of feedback can help make you faster, many drivers prefer the finer control and feel of manual brakes. When the brake pedal ratio and master cylinder bore are appropriately chosen to match the calipers and pad bite, leg effort won't be excessive even compared to a power-assisted unit. For this reason, many drivers prefer non-assisted brakes because they are easier to modulate and threshold brake at the edge of lock-up.

Getting the pedal ratio (and master cylinder bore size) right is critical for a non-assisted brake system to work well and not fatigue your leg in hard driving. It is impossible to give a hard number that works for every vehicle, but you want a pedal ratio of more than 6:1. The size of the master cylinder piston is even harder to pinpoint, but typically it should be less than 1". Researching the size of the factory unit originally fitted with unassisted manual brakes is a good place to start.

Vacuum Boosters



Figuring how much assistance you get from a vacuum booster is simple math. A mildly tuned engine with 16-inches of vacuum at idle translates to about -8 psi (actually 7.86, this is complicated to explain, so [Google the conversion](#)). So if you have -8 psi on one side of a small 7-inch diameter vacuum diaphragm booster (as small as they come) and regular atmosphere on the other, it adds about 300 pounds of assist when you hit the pedal.

To get that number, multiply the diaphragm area in square inches by 8 psi (ignore the negative nature of the number). The surface area of a circle is found with the formula of $A = \pi \times r^2$ (we'll pretend the dead area in the center of the diaphragm with the pushrod isn't there). If it has been a while since high school geometry, that means $3.14 \times \text{half the diameter} \times \text{half the diameter}$. In this case, $3.14 \times 3.5 \times 3.5 = 38.465$ sq. inches $\times 8$ psi = 307.7 pounds of force.

A larger diameter booster gives more boost; moving up to an 8-inch size adds an extra 100-pounds. The math works like this, $4 \times 4 \times 3.14 = 50.24$ sq. inches $\times 8$ psi = 401.9 pounds of force. A dual-diaphragm booster doubles the area and the boost.

But you need vacuum to use any booster, and you need room to mount it. If the only issue is a lack of sufficient vacuum from the engine, a vacuum pump may be the simplest solution. Some vehicles used a motor-driven vacuum pump mounted to the rear of the alternator from the factory. There are also electric vacuum pumps available.

Hydro boost



Another solution is to use a hydraulic assist, or Hydro-boost, system. These units can be found in the junkyards or purchased new from aftermarket suppliers. GM used hydraulic brake boosters between the master cylinder and firewall on many 3/4 and 1-ton trucks (and the Buick Grand National), and Ford used one on the 1999-2004 Mustang because of a lack of space.

Hydro-boost (Bendix's name for the system when invented in 1973) uses pressurized fluid from the power steering pump instead of engine vacuum to amplify your leg effort. Of course, you need a power steering pump to make it work, but it is just a matter of plumbing if you already have power steering.

How much boost you get is dependent on how much pressure the power steering pump puts out, but more than 1000 pounds is typical. Hydro-boost works the same no matter the throttle position or engine load, unlike vacuum.

Off-road trucks and buggies have adopted Hydro-boost because it allows drivers to control low-speed crawling with oversized tires. Just a little effort on the pedal can deliver 1500 pounds of force to the master cylinder.

Conclusion

There is a reason that all the automakers stick with vacuum-assisted power brakes – it is an affordable and mostly trouble-free system. For most people building most cars it is probably the default option. An

auxiliary pump can still make this a viable option if you don't get quite as much engine vacuum as you need.

For power brakes for your big tire off-road truck, with really low gearing and lots of torque, Hydro-boost is an increasingly popular option. Hydro-boost can also free up some room under the hood for a big block or OHC V8.

But Wilwood does sell many unassisted braking systems to racers in the top classes around the world. Nothing beats full manual when you are trying to modulate the brakes for that last sliver of competitive edge. If you figure your pedal ratio correctly and use the appropriate bore size master cylinder, you can have manual brakes that stop with authority without undue leg effort.

If you need more technical help figuring out your brake system, [email the Wilwood tech staff](#).

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